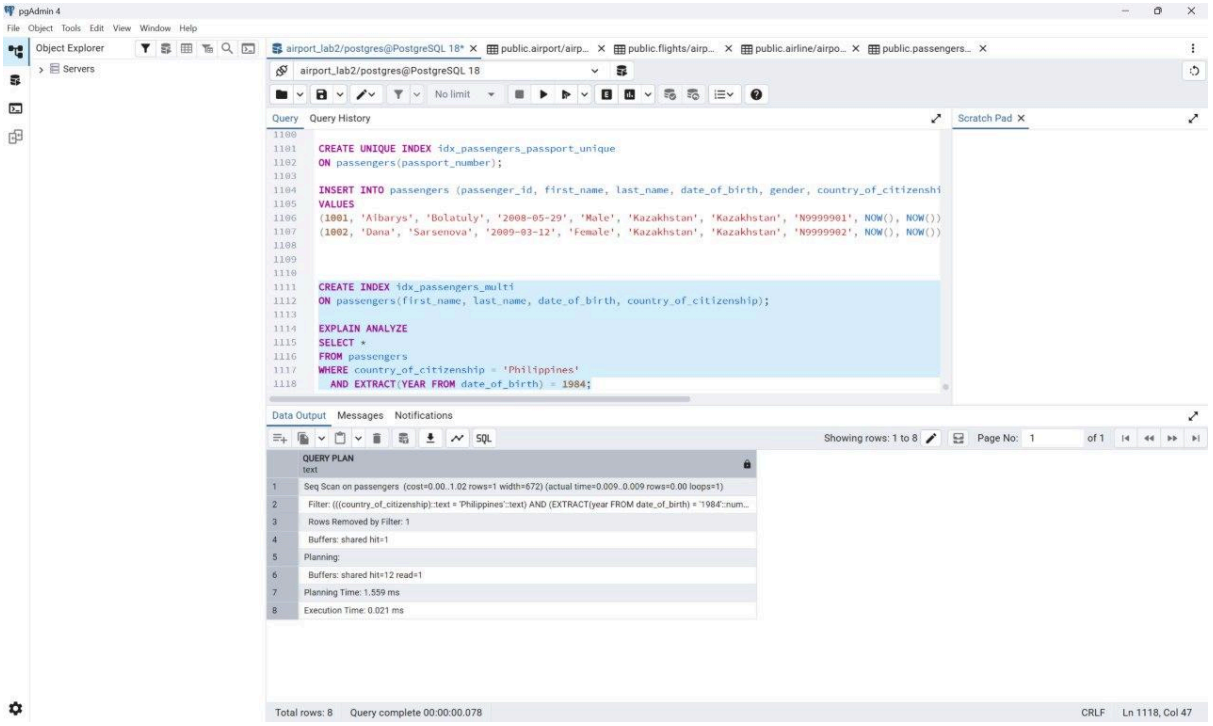
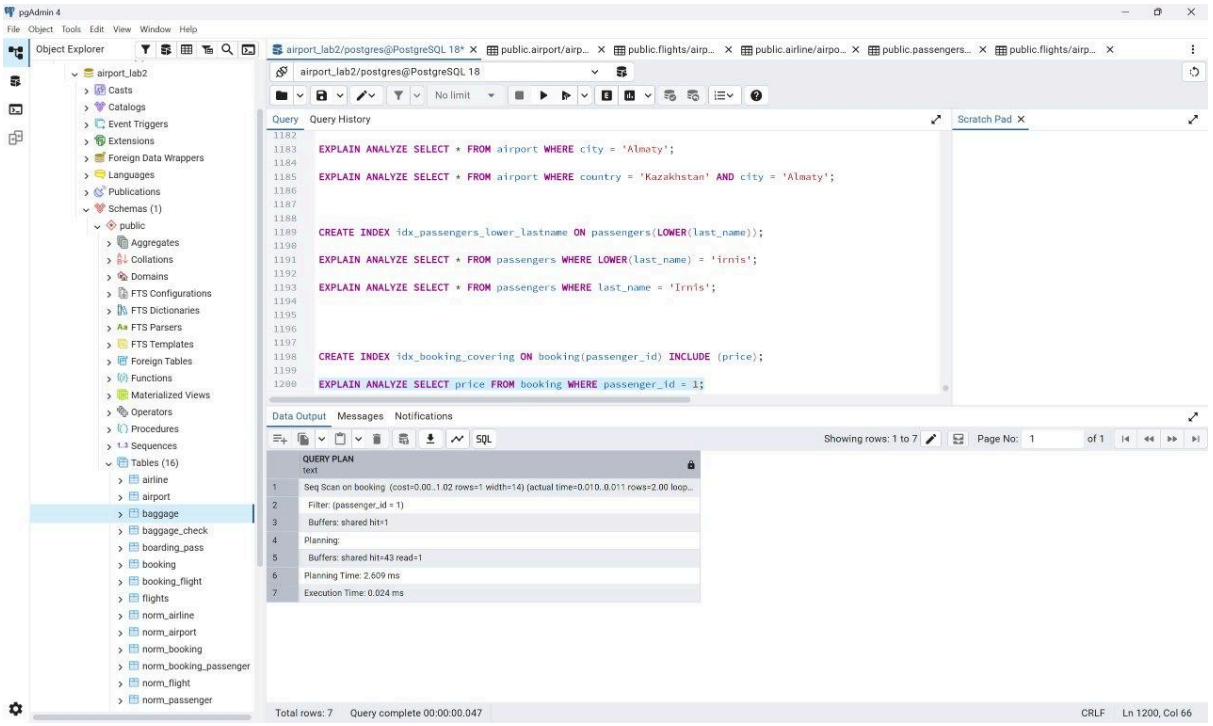
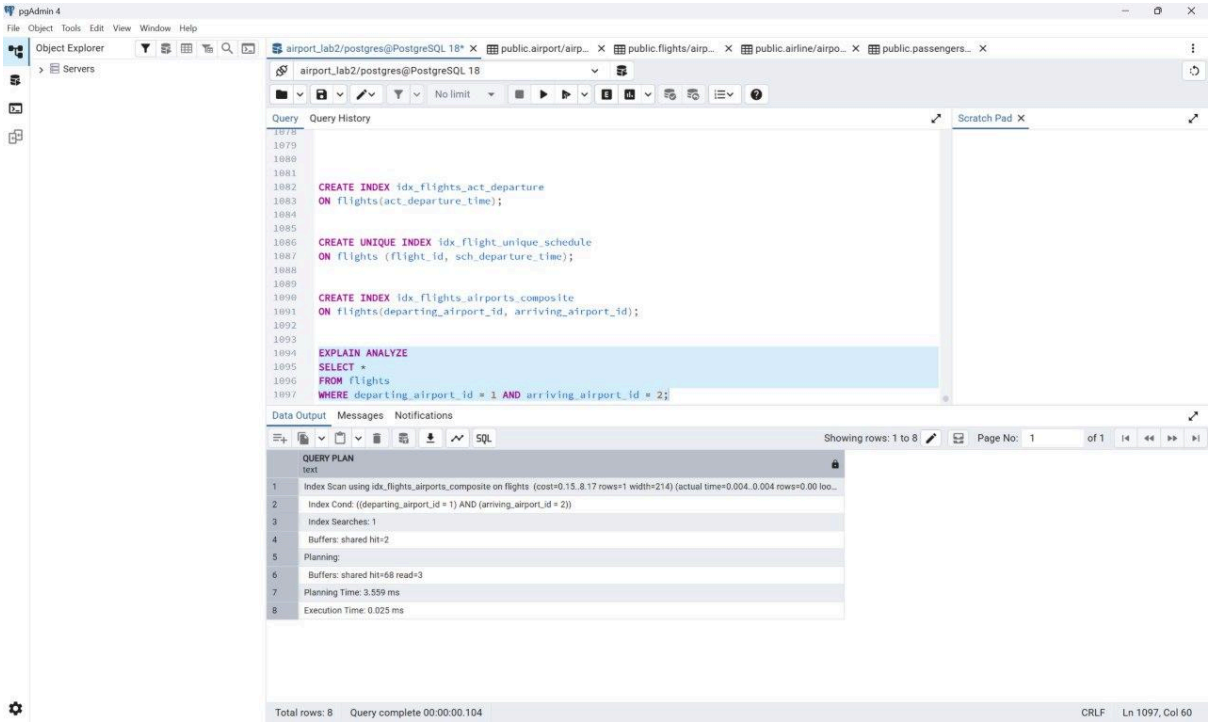
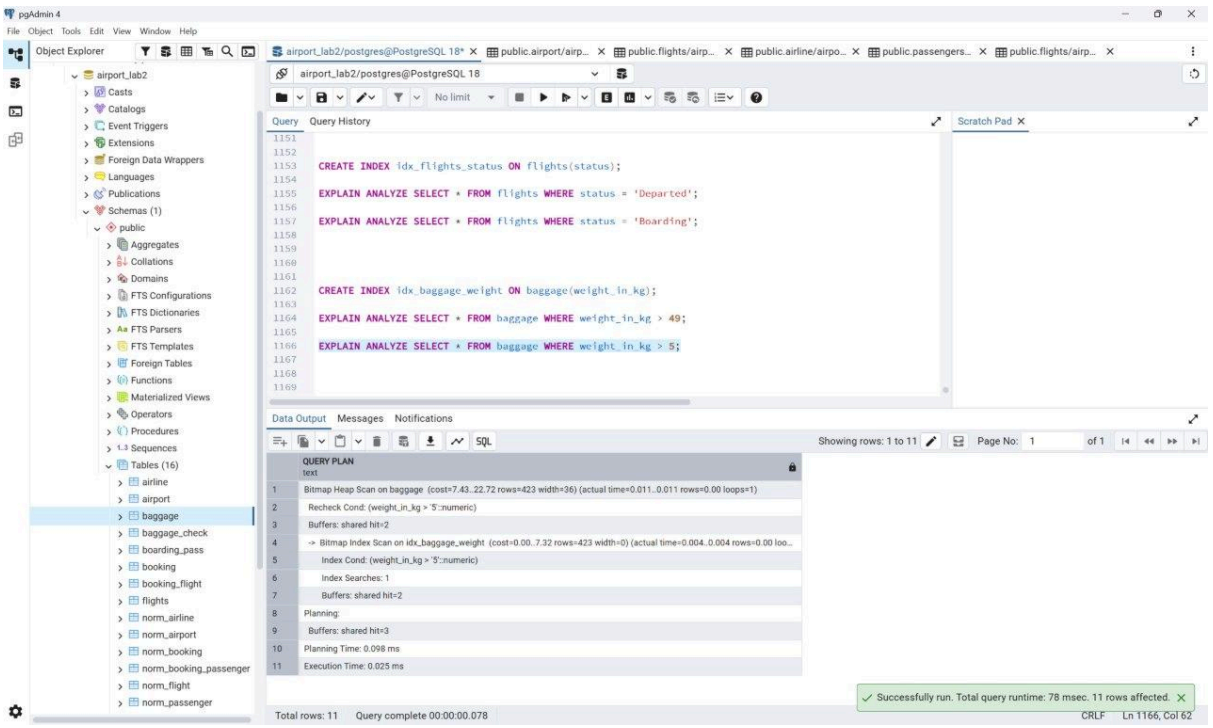
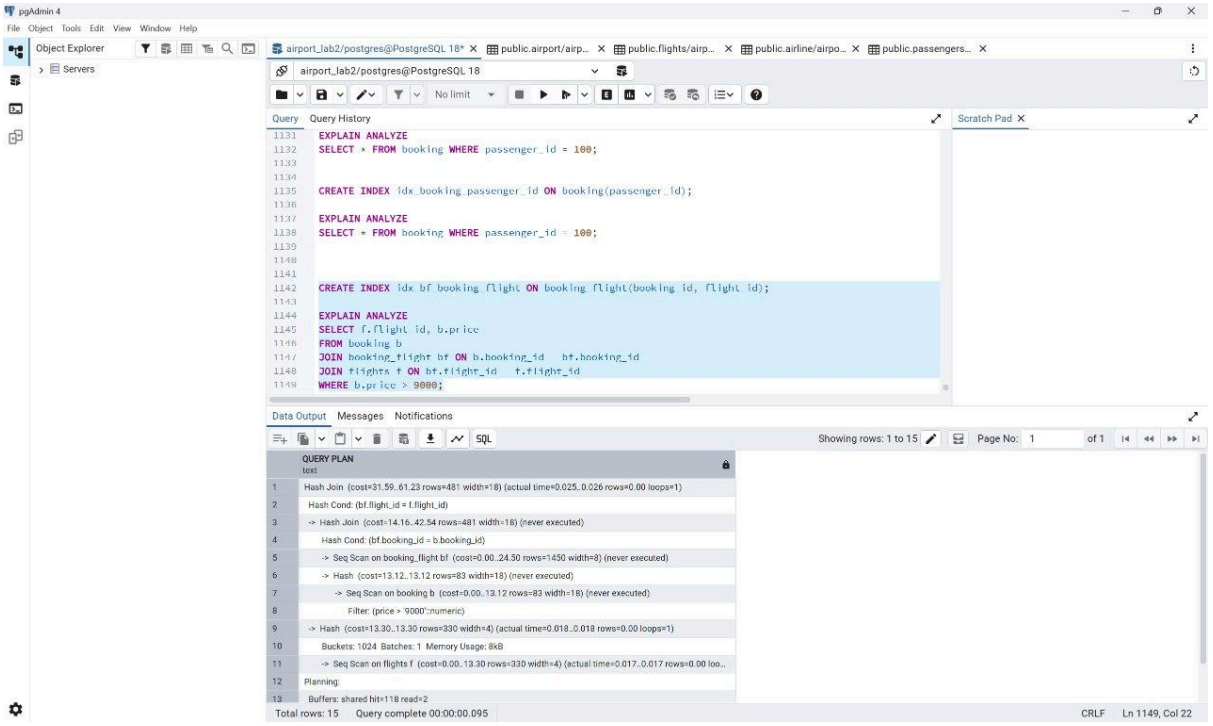
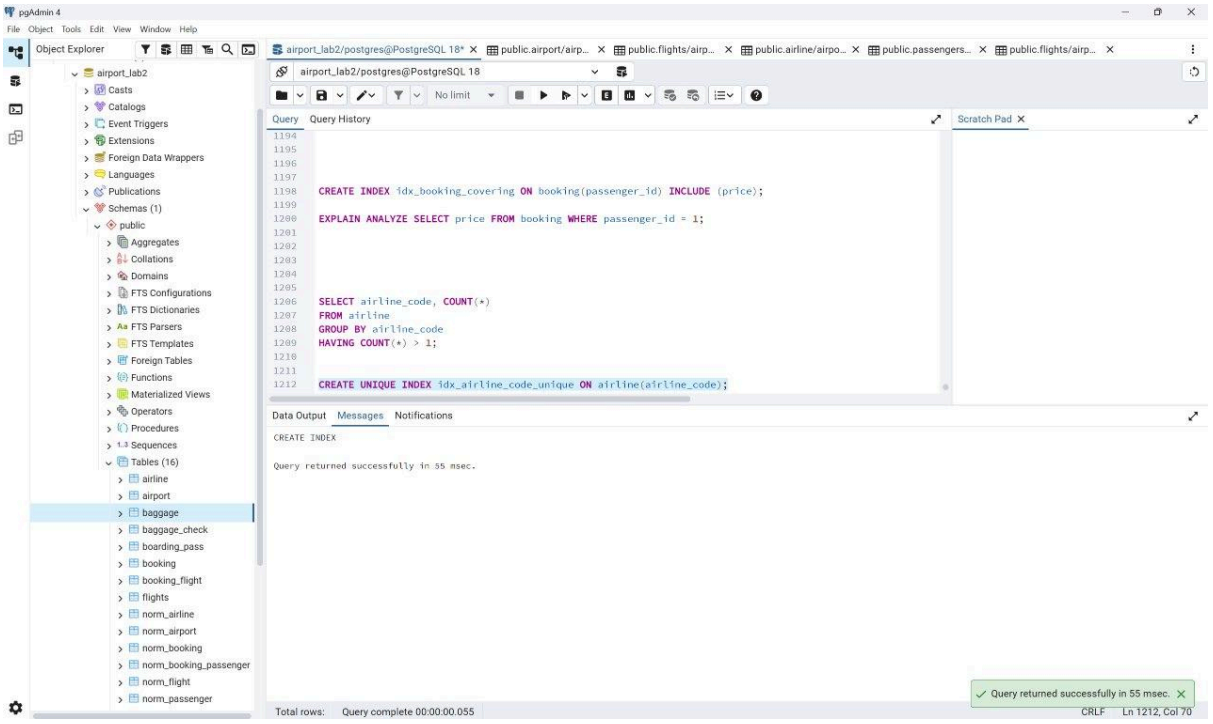

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public.airport/airp...
public.flights/airp...
public.airline/airpo...
public.passengers...
airport_lab2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1100
1101 CREATE UNIQUE INDEX idx_passengers_passport_unique
1102 ON passengers(passport_number);
1103
1104 INSERT INTO passengers (passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenshi
1105 VALUES
1106 (1001, 'Albarys', 'Bolatuly', '2008-05-29', 'Male', 'Kazakhstan', 'Kazakhstan', 'N9999901', NOW(), NOW())
1107 (1002, 'Dana', 'Sarsenova', '2009-03-12', 'Female', 'Kazakhstan', 'Kazakhstan', 'N9999902', NOW(), NOW())
1108
1109
1110
1111 CREATE INDEX idx_passengers_multi
1112 ON passengers(first_name, last_name, date_of_birth, country_of_citizenship);
1113
1114 EXPLAIN ANALYZE
1115 SELECT *
1116 FROM passengers
1117 WHERE country_of_citizenship = 'Philippines'
1118 AND EXTRACT(YEAR FROM date_of_birth) = 1984;
Data Output
Messages
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SQL
Showing rows: 1 to 8
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QUERY PLAN
text
1 Seq Scan on passengers (cost=0.00..1.02 rows=1 width=672) (actual time=0.009..0.009 rows=0.00 loops=1)
2 Filter: (((country_of_citizenship)::text = 'Philippines'::text) AND (EXTRACT(year FROM date_of_birth) = '1984'::num...
3 Rows Removed by Filter: 1
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=12 read=1
7 Planning Time: 1.559 ms
8 Execution Time: 0.021 ms
Total rows: 8
Query complete 00:00:00.078
CRLF
Ln 1118, Col 47

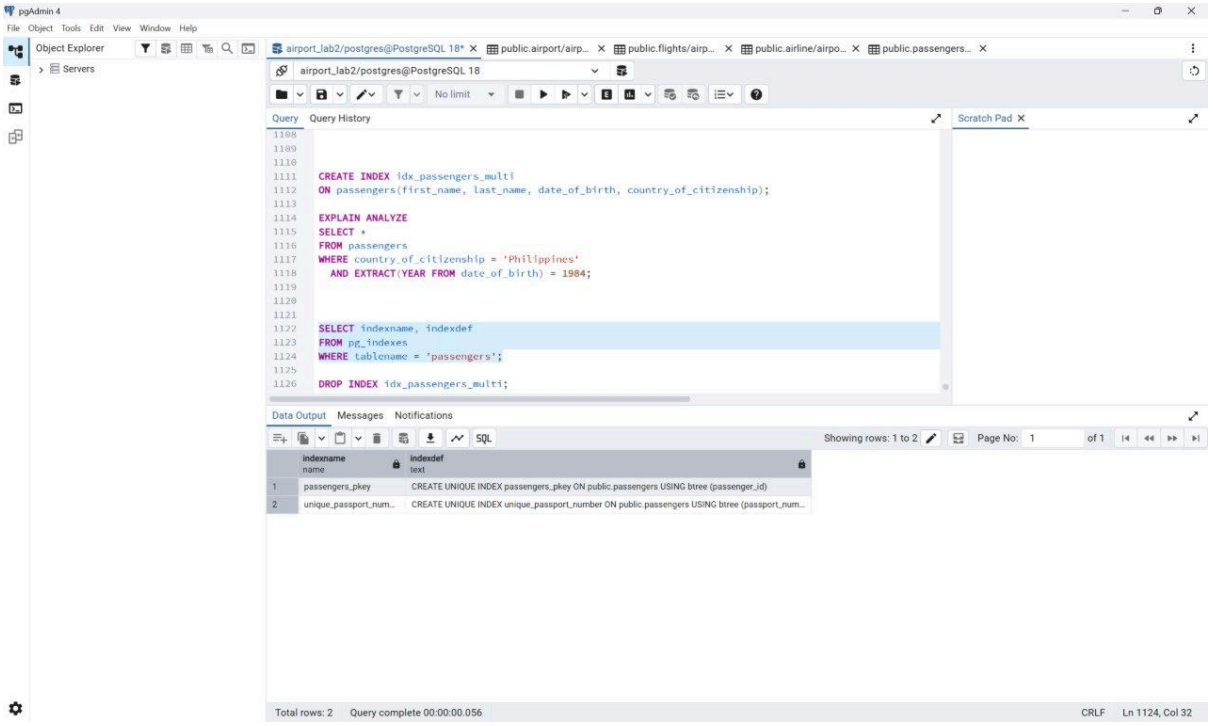

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baggage_check
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booking
booking_flight
flights
norm_airline
norm_airport
norm_booking
norm_booking_passenger
norm_flight
norm_passenger
airport_lab2/postgres@PostgreSQL 18*
public.airport/airp...
public.flights/airp...
public.airline/airpo...
public.passengers...
public.flights/airp...
airport_lab2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1182
1183 EXPLAIN ANALYZE SELECT * FROM airport WHERE city = 'Almaty';
1184
1185 EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Kazakhstan' AND city = 'Almaty';
1186
1187
1188
1189 CREATE INDEX idx_passengers_lower_lastname ON passengers(LOWER(last_name));
1190
1191 EXPLAIN ANALYZE SELECT * FROM passengers WHERE LOWER(last_name) = 'irnis';
1192
1193 EXPLAIN ANALYZE SELECT * FROM passengers WHERE last_name = 'Irnis';
1194
1195
1196
1197
1198 CREATE INDEX idx_booking_covering ON booking(passenger_id) INCLUDE (price);
1199
1200 EXPLAIN ANALYZE SELECT price FROM booking WHERE passenger_id = 1;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 7
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on booking (cost=0.00..1.02 rows=1 width=14) (actual time=0.010..0.011 rows=2.00 loop...
2 Filter: (passenger_id = 1)
3 Buffers: shared hit=1
4 Planning:
5 Buffers: shared hit=43 read=1
6 Planning Time: 2.609 ms
7 Execution Time: 0.024 ms
Total rows: 7
Query complete 00:00:00.047
CRLF
Ln 1200, Col 66


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public.flights/airp...
public.airline/airpo...
public.passengers...
airport_lab2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1078
1079
1080
1081
1082 CREATE INDEX idx_flights_act_departure
1083 ON flights(act_departure_time);
1084
1085
1086 CREATE UNIQUE INDEX idx_flight_unique_schedule
1087 ON flights (flight_id, sch_departure_time);
1088
1089
1090 CREATE INDEX idx_flights_airports_composite
1091 ON flights(departing_airport_id, arriving_airport_id);
1092
1093
1094 EXPLAIN ANALYZE
1095 SELECT *
1096 FROM flights
1097 WHERE departing_airport_id = 1 AND arriving_airport_id = 2;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Index Scan using idx_flights_airports_composite on flights (cost=0.15..8.17 rows=1 width=214) (actual time=0.004..0.004 rows=0.00 loo...
2 Index Cond: ((departing_airport_id = 1) AND (arriving_airport_id = 2))
3 Index Searches: 1
4 Buffers: shared hit=2
5 Planning:
6 Buffers: shared hit=68 read=3
7 Planning Time: 3.559 ms
8 Execution Time: 0.025 ms
Total rows: 8
Query complete 00:00:00.104
CRLF
Ln 1097, Col 60


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airline
airport
baggage
baggage_check
boarding_pass
booking
booking_flight
flights
norm_airline
norm_airport
norm_booking
norm_booking_passenger
norm_flight
norm_passenger
airport_lab2/postgres@PostgreSQL 18*
public.airport/airp...
public.flights/airp...
public.airline/airpo...
public.passengers...
public.flights/airp...
airport_lab2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1151
1152
1153 CREATE INDEX idx_flights_status ON flights(status);
1154
1155 EXPLAIN ANALYZE SELECT * FROM flights WHERE status = 'Departed';
1156
1157 EXPLAIN ANALYZE SELECT * FROM flights WHERE status = 'Boarding';
1158
1159
1160
1161
1162 CREATE INDEX idx_baggage_weight ON baggage(weight_in_kg);
1163
1164 EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg > 49;
1165
1166 EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg > 5;
1167
1168
1169
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 11
Page No: 1
of 1
QUERY PLAN
text
1 Bitmap Heap Scan on baggage (cost=7.43..22.72 rows=423 width=36) (actual time=0.011..0.011 rows=0.00 loops=1)
2 Recheck Cond: (weight_in_kg > '5'::numeric)
3 Buffers: shared hit=2
4 -> Bitmap Index Scan on idx_baggage_weight (cost=0.00..7.32 rows=423 width=0) (actual time=0.004..0.004 rows=0.00 loo...
5 Index Cond: (weight_in_kg > '5'::numeric)
6 Index Searches: 1
7 Buffers: shared hit=2
8 Planning:
9 Buffers: shared hit=3
10 Planning Time: 0.098 ms
11 Execution Time: 0.025 ms
Successfully run. Total query runtime: 78 msec. 11 rows affected.
Total rows: 11
Query complete 00:00:00.078
CRLF
Ln 1166, Col 62


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public.flights/airp...
public.airline/airpo...
public.passengers...
airport_lab2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1131 EXPLAIN ANALYZE
1132 SELECT * FROM booking WHERE passenger_id = 100;
1135 CREATE INDEX idx_booking_passenger_id ON booking(passenger_id);
1137 EXPLAIN ANALYZE
1138 SELECT * FROM booking WHERE passenger_id = 100;
1142 CREATE INDEX idx_bf_booking_flight ON booking_flight(booking_id, flight_id);
1144 EXPLAIN ANALYZE
1145 SELECT f.flight_id, b.price
1146 FROM booking b
1147 JOIN booking_flight bf ON b.booking_id = bf.booking_id
1148 JOIN flights f ON bf.flight_id = f.flight_id
1149 WHERE b.price > 9000;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 15
Page No: 1
of 1
QUERY PLAN
text
1 Hash Join (cost=31.59..61.23 rows=481 width=18) (actual time=0.025..0.026 rows=0.00 loops=1)
2 Hash Cond: (bf.flight_id = f.flight_id)
3 -> Hash Join (cost=14.16..42.54 rows=481 width=18) (never executed)
4 Hash Cond: (bf.booking_id = b.booking_id)
5 -> Seq Scan on booking_flight bf (cost=0.00..24.50 rows=1450 width=8) (never executed)
6 -> Hash (cost=13.12..13.12 rows=83 width=18) (never executed)
7 -> Seq Scan on booking b (cost=0.00..13.12 rows=83 width=18) (never executed)
8 Filter: (price > '9000'::numeric)
9 -> Hash (cost=13.30..13.30 rows=330 width=4) (actual time=0.018..0.018 rows=0.00 loops=1)
10 Buckets: 1024 Batches: 1 Memory Usage: 8kB
11 -> Seq Scan on flights f (cost=0.00..13.30 rows=330 width=4) (actual time=0.017..0.017 rows=0.00 loo...
12 Planning:
13 Buffers: shared hit=118 read=2
Total rows: 15
Query complete 00:00:00.095
CRLF
Ln 1149, Col 22


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norm_airport
norm_booking
norm_booking_passenger
norm_flight
norm_passenger
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public.flights/airp...
public.airline/airpo...
public.passengers...
public.flights/airp...
airport_lab2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1198 CREATE INDEX idx_booking_covering ON booking(passenger_id) INCLUDE (price);
1200 EXPLAIN ANALYZE SELECT price FROM booking WHERE passenger_id = 1;
1206 SELECT airline_code, COUNT(*)
1207 FROM airline
1208 GROUP BY airline_code
1209 HAVING COUNT(*) > 1;
1212 CREATE UNIQUE INDEX idx_airline_code_unique ON airline(airline_code);
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 55 msec.
Query returned successfully in 55 msec.
Total rows:
Query complete 00:00:00.055
CRLF
Ln 1212, Col 70


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Servers
airport_lab2/postgres@PostgreSQL 18*
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public.flights/airp...
public.airline/airpo...
public.passengers...
airport_lab2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1108
1109
1110
1111 CREATE INDEX idx_passengers_multi
1112 ON passengers(first_name, last_name, date_of_birth, country_of_citizenship);
1113
1114 EXPLAIN ANALYZE
1115 SELECT *
1116 FROM passengers
1117 WHERE country_of_citizenship = 'Philippines'
1118 AND EXTRACT(YEAR FROM date_of_birth) = 1984;
1119
1120
1121
1122 SELECT indexname, indexdef
1123 FROM pg_indexes
1124 WHERE tablename = 'passengers';
1125
1126 DROP INDEX idx_passengers_multi;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 2
Page No: 1
of 1
indexname
name
indexdef
text
1 passengers_pkey CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
2 unique_passport_num... CREATE UNIQUE INDEX unique_passport_number ON public.passengers USING btree (passport_num...
Total rows: 2
Query complete 00:00:00.056
CRLF
Ln 1124, Col 32